Sudhir Kumar Singh

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Summary

Final-year AI & Data Science undergraduate (CGPA 9.01) with two AI internships building production systems. Manufactured a multi-tenant voice agent platform using Gemini, RAG, tool-calling and real-time STT/TTS, reducing latency from 2s to under 1s. Experienced in LLM agents, retrieval, evaluation, Python backends and AWS/GCP deployment.

Technical Skills

LLMs & Agents: Gemini, LLaMA, Ollama, Prompt Engineering, AI Agents, Multi-Agent Systems, Tool-Calling, LangChain, LangGraph

RAG & NLP: RAG (Hybrid Dense + Sparse Search), Embeddings, FAISS, pgvector, Transformers, NLP, PaddleOCR

Voice AI: Sarvam STT/TTS, Silero VAD, Barge-in / Interruption Handling, WebRTC, Real-time Latency Optimization

Evaluation & MLOps: LangSmith, RAGAS, MLflow, Latency Tracing & Logging, Docker, Kubernetes, Git/GitHub

Backend & Cloud: Python, FastAPI, REST APIs, Celery, Redis, PostgreSQL, MongoDB, GCP, AWS EC2 (GPU)

ML / CV & Languages: PyTorch, TensorFlow, Scikit-learn, XGBoost, YOLO26/YOLOv8, OpenCV; SQL, C++, Java, JavaScript/TypeScript

Experience

AI Engineer Intern — Acuron AI Solutions | *Startup* Jul 2026 – Present

Voice & Chat AI Agent Platform

- Built a multi-tenant **agentic AI platform** that lets any organization launch its own voice, chat and WhatsApp agents in **under 5 minutes**, with dynamic prompts, tool-calling and **hybrid dense + sparse RAG** over a **PostgreSQL + pgvector** knowledge base; evaluated with **LangSmith and RAGAS** and deployed on **AWS EC2**.
- Cut voice-agent latency from **2s to under 1s** by profiling the STT → LLM → TTS chain hop by hop, and built natural real-time conversations with **Sarvam STT/TTS, Gemini 2.5 Flash, WebRTC** and **Silero VAD barge-in** so callers can interrupt mid-response.

Helmet-Compliance Detection — Tata Motors (client)

- Trained a **YOLO26s** detector on **100K+ annotated frames** on an AWS RTX 4090 GPU server; now **live in production** at the client site, with augmentation and negative samples to curb false positives.

AI Intern — TANSU AI | *Startup* Feb 2026 – Apr 2026

- Built a **real-time analytics pipeline** over live RTSP streams: **YOLOv8 + OpenCV** face detection, **AWS Rekognition** identity matching and **ByteTrack** tracking with persistent IDs.
- Mapped dining-table zones to measure per-meal seating time and flag new individuals; designed **GCP storage workflows** for face crops and metadata and optimized inference throughput.

Projects

NomadIQ — Multi-Agent LLM Planner | *FastAPI, TypeScript, Ollama, Gemini* GitHub

- Orchestrated **5 specialized LLM agents** (destination, hotel, dining, budget, itinerary) in parallel to produce day-by-day plans, plus a conversational editor to refine any part in plain English.
- Wrote a unified provider layer switching between **local Ollama and cloud Gemini**; streamed async background-task progress to a live frontend.

Smart Traffic Violation Detection with OCR | *YOLOv8, PaddleOCR, FastAPI, Celery, Redis, Docker* GitHub

- Flagged helmet violations, triple riding and plates with a custom **4-class YOLOv8** model; read Indian plates with **PaddleOCR** plus perspective correction and CLAHE, robust to blur, tilt and low light.
- Built an async **FastAPI + Celery + Redis** backend shipped as **4 Docker Compose services** behind a Next.js dashboard.

Education

B.E., Artificial Intelligence & Data Science — Savitribai Phule Pune University Aug 2023 – Jun 2027
CGPA: 9.01 / 10

Certifications

Machine Learning, DL & NLP — Udemy
The Ultimate Job Ready Data Science Course — CodeWithHarry
Full Stack Web Development Certification — Apna College
Postman API Fundamentals Certification — Postman